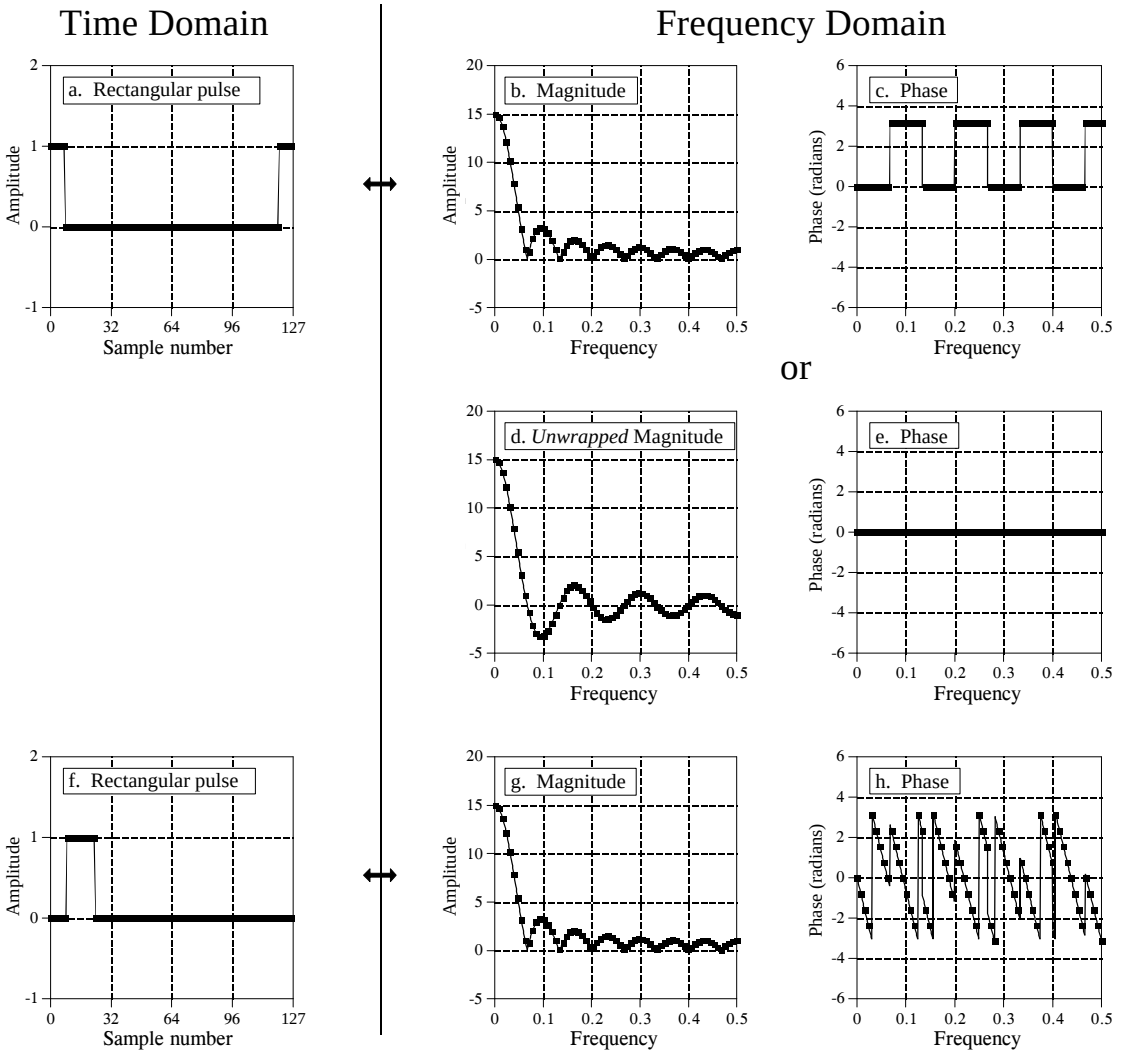



FIGURE 11-3
DFT of a rectangular pulse. A rectangular pulse in one domain corresponds to a sinc function in the other domain.

Alternatively, the DTFT can be used to express the frequency spectrum as a fraction of the sampling rate, *f*:

EQUATION 11-2
Equation 11-1 rewritten in terms of the sampling frequency. The parameter, $f$, is the fraction of the sampling rate, running continiously from 0 to 0.5. To avoid the division by zero, use $Mag\ X(0) = M$.

$$Mag\ X(f) = \left| \frac{\sin(\pi f M)}{\sin(\pi f)} \right|$$

In other words, Eq. 11-1 provides $N/2 + 1$ *samples* in the frequency spectrum, while Eq. 11-2 provides the *continuous curve* that the samples lie on. These